

1.3 Coordinate geometry

Pure Mathematics 1 · 2026 examination · Concept guide and original exam-style practice

Understand the concept

For a line gradient $m = (y_2 - y_1) / (x_2 - x_1)$; parallel lines share gradient and perpendicular non-vertical gradients multiply to -1 . A circle $(x - a)^2 + (y - b)^2 = r^2$ has centre (a, b) . A tangent is perpendicular to radius at contact.

Worked example: follow the reasoning

Circle $x^2 + y^2 - 4x + 6y - 12 = 0$ becomes $(x - 2)^2 + (y + 3)^2 = 25$.

Structured practice

Answer in the space provided. Show working and units where relevant.

1 Find equation of tangent to circle $x^2 + y^2 = 25$ at $(3, 4)$.

[3]

Final answer: _____

Answers and explanation

Question 1 [3 marks]

Radius gradient $\frac{4}{3}$ [1], tangent gradient $-\frac{3}{4}$ [1]; $y-4=-\frac{3}{4}(x-3)$ [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Mathematics 9709, 2026–2027. <https://www.cambridgeinternational.org/Images/697427-2026-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.